

Unit 3 Project Brief

Drone Mission + VEX Ground Support System

Autonomous Aerospace Systems: Controls, Drones & Fluid Power

Challenge Statement: Program and test an autonomous drone mission, then integrate a ground support system that helps the mission operate safely, accurately, or reliably.

Project Overview

In this project, your team programs a Robolink drone mission and designs a VEX, sensor-based, or fluid-power ground support system that connects to the mission environment. The focus is control logic, system behavior, feedback, reliability, and safe operation.

Mission Scenario

Modern aerospace systems combine air vehicles, ground support, sensors, automation, and control logic. A drone may inspect, deliver, survey, or land while a ground system guides, signals, positions, protects, or triggers part of the mission. Your team will create a small-scale version of that interaction.

Design Requirements

- Create a mission objective such as inspection, survey, target landing, obstacle navigation, payload simulation, or checkpoint route.
- Develop a flowchart before programming the mission.
- Program at least one open-loop mission behavior and identify its limitations.
- Add at least one control improvement using sensor feedback, conditional logic, thresholds, repeated testing, or revised timing.
- Design a ground support element such as landing pad, gate, beacon, inspection zone, sensor target, payload platform, VEX actuator, or fluid-power model.
- Test mission performance across repeated trials and document reliability, accuracy, time, or success rate.

Constraints

- All drone operations must follow classroom flight safety rules and approved flight zones.
- Drone code must be tested progressively; unsafe or unpredictable code may not be flown.
- Ground support system must not contact propellers or create unsafe flight obstacles.
- VEX or fluid-power components must be stable and safe before they interact with a drone mission.
- Final mission must be documented with flowchart, code evidence, test data, and system diagram.

Required Engineering Evidence

Control Logic	Flowchart showing sequence, decisions, inputs, outputs, and feedback points.
Programming Evidence	Code screenshots or file evidence with comments explaining key mission logic.
System Integration	Diagram showing the drone, ground support system, inputs, outputs, sensors, and interactions.
Testing	Repeated trials measuring mission success, accuracy, time, reliability, or error.

Improvement	Explanation of how feedback, code revision, or ground support improved performance.
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Project Checkpoints

- 1 Mission objective approved
- 2 Flowchart complete
- 3 Open-loop test completed
- 4 Ground system integrated
- 5 Final mission test and review delivered

Success Criteria

Mission Function	Drone and ground support system complete the intended mission safely and repeatably.
Control System Thinking	Flowchart, code, sensors, feedback, and logic are clearly connected.
Integration	Ground support system meaningfully supports or interacts with the mission.
Testing & Debugging	Performance data and revisions show thoughtful debugging and improvement.

Final Submission

Submit a complete design package that includes your notebook evidence, sketches or CAD evidence, prototype/build evidence, test data, calculations or analysis, and a final design review presentation. Your final recommendation should clearly explain what your design does, how well it performs, and what should be improved next.